

A diagonal-weight influence ranked studies correctly only with two-arm studies and heterogeneity held fixed

A simulation study answering catalog problem CMP-24

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Abstract

Background. `cpaic`'s `edge_influence()` reports each study's contribution to a requested contrast from a hat matrix with diagonal inverse-variance weights and $\hat{\tau}^2$ held fixed. The catalog (CMP-24) says this is exact for two-arm studies at fixed weights and approximate otherwise. The refuting sentence: the approximation preserves the ranking even where it misstates values.

Methods. Random connected four-treatment networks of 8 or 16 studies, all two-arm or with three-arm studies, true τ of 0, 0.1 or 0.3; 500 networks per cell. The diagonal influence was compared with leave-one-study-out refits of the correct model (within-study covariance, τ re-estimated by REML), scored by variance importance. Registered rule: not fit for ranking if in any cell the mean Spearman correlation is below 0.9 or the most important study is misidentified in more than 10% of networks.

Results. The diagonal influence equaled the correct hat row for two-arm networks (largest difference 8.2e-15) and ranked studies well there with τ fixed at 0 (Spearman 0.95 to 0.97, top study missed in 5% to 6%). With three-arm studies and τ fixed, Spearman stayed at 0.90 to 0.92 but the top study was missed in 28% to 46% of networks, almost always because a three-arm study was promoted. With τ re-estimated, the mean Spearman was 0.51 to 0.70 (two-arm networks 0.53 to 0.70) and the top study was missed in 20% to 59%. The registered verdict is **not fit for ranking**.

Conclusion. The refuting sentence fails. Both dropped terms change which study the diagnostic names as most influential; the heterogeneity term lowers the rank correlation more.

1 The problem

An influence diagnostic is read as an ordering: which study carries the contrast. `cpaic`'s calculation writes the contrast as $a'y$ over all pairwise rows, $a = m'(X'WX)^+X'W$ with $W = \text{diag}\{1/(v_j + \hat{\tau}^2)\}$, and sums $|a_j|$ over each study's rows. Two terms are dropped. A multi-arm study's pairwise rows are correlated and linearly dependent, so the diagonal weights count its information more than once. Removing a study changes $\hat{\tau}^2$ and thereby every weight, which a fixed- $\hat{\tau}$ hat matrix cannot see. The reference is the refitted influence, following the variance-based definition of study importance (1).

2 Design

Registered protocol: `protocol.md`. Treatments A to D with true effects (0, -0.2, -0.3, -0.4) against A; target D versus A. Each study compares a random pair, or with probability 0.3 a random triple (three-arm cells); arm sizes 50 to 300; random effects with correlation 0.5 within multi-arm studies. The diagonal influence is reimplemented from `cpaic`'s formula (commit cf27b1a) with $\hat{\tau}^2$ from the correct model. Reference: leave-one-study-out refits of the correct model (basic contrasts, block within-study covariance, REML τ^2 ; τ

fixed at 0 in the $\tau = 0$ cells), scored by variance importance $1 - \text{Var}_{\text{full}} / \text{Var}_s$ and by absolute change in the estimate. The $\tau = 0$ cells therefore isolate the multi-arm term; the $\tau > 0$ cells add the re-estimation term. Networks in which removing one study disconnected a treatment other than A or D made the refit singular and were dropped: 120 of 6000, all in 8-study cells (2% to 6% per cell).

3 Results

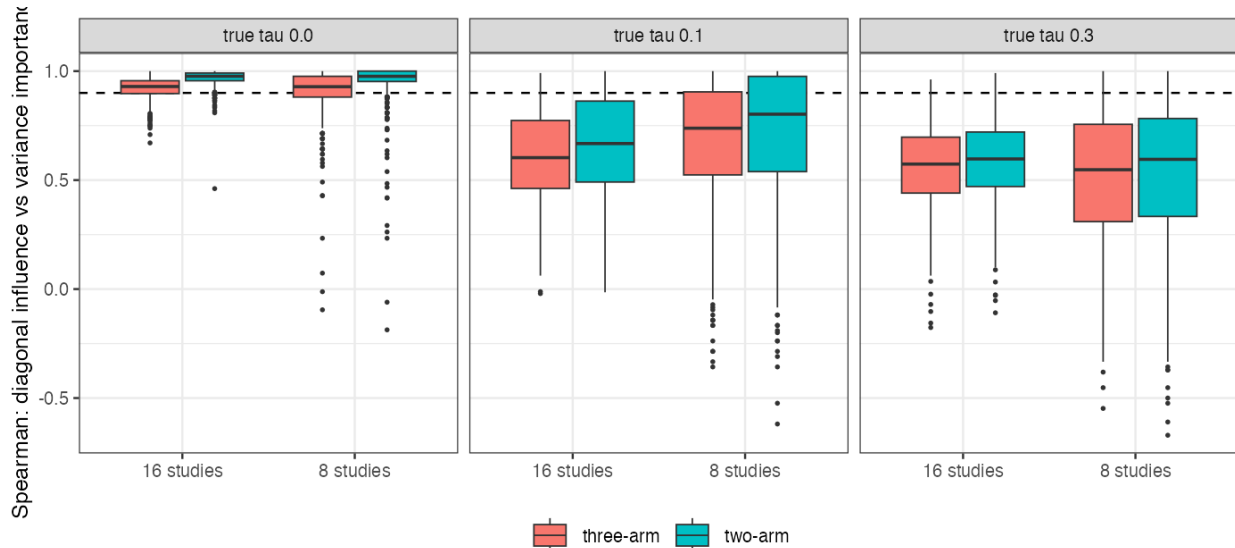


Figure 1: Per-network Spearman correlation between the diagonal influence and refitted variance importance; dashed line, the registered 0.9.

Table 1: Registered results. Spearman correlations are means over networks.

studies	design	true τ	networks	mean Spearman	share below 0.9	top study missed	Spearman with estimate change
8	two-arm	0.0	480	0.95	0.10	0.05	0.58
8	three-arm	0.0	484	0.90	0.29	0.28	0.55
8	two-arm	0.1	471	0.70	0.61	0.20	0.59
8	three-arm	0.1	487	0.66	0.74	0.37	0.55
8	two-arm	0.3	468	0.53	0.89	0.34	0.55
8	three-arm	0.3	490	0.51	0.92	0.47	0.51
16	two-arm	0.0	500	0.97	0.05	0.06	0.57
16	three-arm	0.0	500	0.92	0.28	0.46	0.51
16	two-arm	0.1	500	0.66	0.78	0.25	0.54
16	three-arm	0.1	500	0.61	0.90	0.51	0.51
16	two-arm	0.3	500	0.58	0.98	0.37	0.54

studies	design	true τ	networks	mean Spearman	share below 0.9	top study missed	Spearman with estimate change
16	three- arm	0.3	500	0.56	0.99	0.59	0.50

Exactness (P1). In every two-arm network the diagonal influence equaled the correct model’s hat row at the same $\hat{\tau}$ to 8.2e-15.

Multi-arm term. With τ fixed at 0, three-arm networks kept a mean Spearman of 0.90 to 0.92, but the most important study was misidentified in 28% to 46% of networks (two-arm: 5% to 6%). A post hoc rerun of the first 150 networks of these cells (`R/05-ties.R`) showed the misses were not near-ties: the reference importance of the study the diagonal named was a median 15% to 21% below the reference maximum. When the two disagreed, the diagonal’s top study was three-arm in 98% to 100% of networks against 18% to 25% for the reference’s, while 30% to 32% of studies were three-arm. Summing absolute coefficients over three dependent rows promotes multi-arm studies, as the design predicted.

Heterogeneity term. With τ re-estimated, the mean Spearman was 0.51 to 0.70 and the top study was missed in 20% to 59%, including two-arm networks where the fixed- $\hat{\tau}$ calculation is exact. Removing a study moves $\hat{\tau}^2$ and so reweights all others; removing a study that raises $\hat{\tau}$ can make the estimate more precise, which gives negative importance. In 8-study two-arm networks with $\tau = 0.3$, 259 of 365 studies with zero diagonal influence (their comparison did not reach D versus A) moved the estimate when removed, through $\hat{\tau}$ alone.

Estimate change. Spearman with the absolute change in the estimate was 0.50 to 0.59 in every cell, including the exact case: the influence measures leverage, while the realized change also depends on the study’s residual.

4 What this does not answer

Standard network meta-analysis, not component networks with population adjustment; study-level, not edge-level, deletion; the formula was reimplemented rather than called, because the package’s working tree was changing during the study. No fixed- $\hat{\tau}$ reference was computed, so in the $\tau > 0$ cells the reference’s own sampling noise through $\hat{\tau}$ is part of the disagreement, which is the dropped term by construction. Whether a block-weighted hat matrix, exact at fixed $\hat{\tau}$ for multi-arm studies, restores the ranking in the $\tau = 0$ cells was not tested. The tie analysis is post hoc. Dropped networks with singular refits are the ones containing a sole bridge to an off-target treatment. Peer review has not been done. Conflict of interest: cpaic is written by this catalog’s author.

References

1. Gerta Rücker, Adriani Nikolakopoulou, Theodoros Papakonstantinou, Georgia Salanti, Richard D. Riley, Guido Schwarzer. The statistical importance of a study for a network meta-analysis estimate. BMC Medical Research Methodology. 2020;20:190. doi:[10.1186/s12874-020-01075-y](https://doi.org/10.1186/s12874-020-01075-y)